Amara Dating App — Notification & Messaging API Specification

Complete Backend & Frontend Contracts • Glambu Interaction Model

Summary for Backend Team: This document outlines the exact endpoints, request schemas, 200 OK JSON responses, and Firebase Cloud Messaging (FCM) push payloads required by the mobile app. All endpoints match existing frontend React Query hooks and the Swagger specification.

1. Architecture & Overview Matrix

Notification Type	HTTP Endpoint / Trigger	Who Triggers It?	Action / Destination
Device Token	`POST /notification/push`	Mobile App (on Login)	Stores user FCM token in database
In-App List	`GET /notification?userId={id}`	Mobile App (Bell icon)	Renders notification activity list
Mark Read	`PUT /notification/read/{id}`	Mobile App (on item tap)	Updates notification read status in DB
Preferences	`GET & PUT /notification/preferences`	Mobile App (Settings screen)	Controls Email & App notification toggles
New Proposal Push	Firebase Admin SDK (FCM)	Backend (`/connections/send`)	Pushes banner to Woman ➡ Opens `Invitations (Pending)`
Accepted Match Push	Firebase Admin SDK (FCM)	Backend (`/connections/accept`)	Pushes banner to Man ➡ Opens `Sent (Accepted)`

2. API Endpoints & Response Payloads

POST `/notification/push`

Registers or updates the mobile device's FCM push token for a user.

REQUEST BODY (APPLICATION/JSON)

```
{
  "userId": "101",
  "token": "dK1s_eXample_FCM_Token_89234jsd98234kjhsdf...",
  "deviceToken": "dK1s_eXample_FCM_Token_89234jsd98234kjhsdf...",
  "platform": "android"
}
```

RESPONSE BODY (200 OK)

```
{
  "status": "success",
  "message": "Device token registered successfully",
  "data": {
    "userId": "101",
    "token": "dK1s_eXample_FCM_Token_89234jsd98234kjhsdf...",
    "platform": "android"
  }
}
```

GET /notification?userId={userId}

Fetches all in-app notifications for the given user. Powers the Bell notification screen.

RESPONSE BODY (200 OK)


```
[
  {
    "id": 1,
    "userId": "101",
    "title": "  Proposal Accepted!",
    "message": "Elena accepted your invitation! You can now chat.",
    "type": "MATCH",
    "read": false,
    "createdAt": "2026-11-03T07:15:30.000Z"
  },
  {
    "id": 2,
    "userId": "101",
    "title": "  New Date Proposal!",
    "message": "Alex sent you an invitation for dinner this Friday. Tap to review.",
    "type": "INVITE",
    "read": true,
    "createdAt": "2026-11-02T18:40:00.000Z"
  },
  {
    "id": 3,
    "userId": "101",
    "title": "  Telegram Available",
    "message": "Telegram username for Priya is now available to view.",
    "type": "TELEGRAM",
    "read": true,
    "createdAt": "2026-11-01T15:20:00.000Z"
  },
  {
    "id": 4,
    "userId": "101",
    "title": "★ Plan Activated",
    "message": "Your 30-day Premium membership is now active.",
    "type": "SYSTEM",
    "read": true,
    "createdAt": "2026-11-01T12:00:00.000Z"
  }
]
```


Note on "type": `MATCH` deep-links men to *Sent (Accepted)*. `INVITE` deep-links women to *Invitations (Pending)*.

PUT /notification/read/{notificationId}

Marks a specific notification item as read when tapped by the user.

QUERY PARAMETERS

```
notificationId=1&userId=101
```

RESPONSE BODY (200 OK)


```
{
  "status": "success",
  "message": "Notification marked as read",
  "notificationId": 1
}
```


GET /notification/preferences?userId={userId}

Retrieves user's notification preferences for Email and App push.

RESPONSE BODY (200 OK)


```
{
  "userId": "101",
  "emailNotifications": true,
  "appNotifications": true
}
```

PUT `/notification/preferences`

Updates user's notification preferences when toggles are switched.

REQUEST BODY (APPLICATION/JSON)

```
{
  "userId": "101",
  "emailNotifications": true,
  "appNotifications": false
}
```

RESPONSE BODY (200 OK)

```
{
  "status": "success",
  "message": "Notification preferences updated successfully",
  "data": {
    "userId": "101",
    "emailNotifications": true,
    "appNotifications": false
  }
}
```

3. Firebase Cloud Messaging (FCM) Push Payloads

The backend must send these messages via the **Firebase Admin SDK (HTTP v1)** to the recipient user's device token.

FCM PUSH When Man Sends Proposal (Delivered to Woman)

Trigger point: Inside `POST /connections/send` handler on the backend.

FCM MESSAGE OBJECT

```
{
  "message": {
    "token": "WOMAN_USER_FCM_DEVICE_TOKEN",
    "notification": {
      "title": "   New Date Proposal!",
      "body": "Alex sent you an invitation for dinner. Tap to review."
    },
    "data": {
      "type": "INVITE",
      "screen": "Invitations",
      "tab": "pending",
      "requestId": "501"
    },
    "android": {
      "priority": "high",
      "notification": {
        "channel_id": "amara_default_channel",
        "sound": "default"
      }
    }
  }
}
```

FCM PUSH When Woman Accepts Proposal (Delivered to Man)

Trigger point: Inside `PUT /connections/accept` handler on the backend.

FCM MESSAGE OBJECT

```
{
  "message": {
    "token": "MAN_USER_FCM_DEVICE_TOKEN",
    "notification": {
      "title": "   Proposal Accepted!",
      "body": "Elena accepted your invitation! You can now chat."
    },
    "data": {
      "type": "MATCH",
      "screen": "Sent",
      "tab": "accepted",
      "requestId": "501"
    },
    "android": {
      "priority": "high",
      "notification": {
        "channel_id": "amara_default_channel",
        "sound": "default"
      }
    }
  }
}
```

4. Transactional Email Payload

Trigger point: Inside `PUT /connections/accept`, send if `user.email_notifications == true`.

Recipient	man@example.com
Subject	It's a Match! Elena accepted your proposal on Amara
Body Text	Hi Alex, Great news! Elena just accepted your invitation for dinner. Open the Amara app to unlock her Telegram handle and start talking. — **The Amara Team**